IFA BORU

SPECIAL BOARDING SECONDARY SCHOOL

Thermal Physics Worksheet
Prepared For Summer Program

BY:
Wondimu Getachew
(Summer, 2016 E.C)
Asella, Ethiopia

Follow Us On:
You Tube: http://www.youtube.com/@WondexAcademy
Telegram: https://t.me/WondexAcademy

1. A gas in a cylinder expanded and does 150J of work in pushing up the piston while 500J of heat is supplied to it. What is the change in the internal energy of the gas in the process?
   A) 150 J B) 650 J C) 500 J D) 350 J

2. Cities have higher average evening temperature than the surrounding countryside. This is because concrete (common building material for roads, buildings, etc in cities) has a higher specific heat capacity than soil, this implies that, in the evening

   A) concrete releases the large amount of energy stored in t during the day and cools faster than soil
   B) soil releases the large amount of energy stored in it during the day and cools slower than concrete
   C) concrete releases the large amount of energy stored in it during the day and cools slower than soil.
   D) soil absorbs large amount of energy and cools slower than concrete

3. Why is an ice at 0^0C more effective than the same mass of water at 0^0C in cooling a drink?
   A) the ice more average kinetic energy than the water
   B) the ice absorbs heat from the drink in order to melt to water
   C) the water transfers heat to the drink in order to freeze to ice
   D) the water particles on the average move faster than the ice.

4. Which of the following systems can absorb some amount of heat without any change in its temperature?
   A) Ice at 0^0C B) water 10^0C
   C) water 0^0C D) ice at -10^0C

5. What is the ratio of the coefficient of linear expansion to coefficient of volume expansion?
   A) 1:2 B) 1:4 C) 1:3 D) 1:6

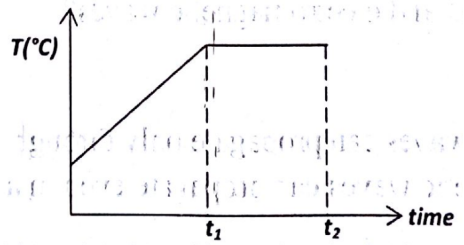
6. Body x and Body y are separately in thermal equilibrium with Body z. Based on these facts, what can you conclude about Body x and Body y?
   A) they have the same temperature.
   B) they have the same heat capacity.
   C) they are in thermal contact.
   D) they are in thermal equilibrium with any other body.

7. Coefficient of expansion of water is greater than that of glass. Two identical glass beakers are full of water at a temperature of 20^0C. Beaker-1 is first warmed to a temperature of 35^0C and then cooled back to a temperature of 20^0C. Beaker-2 is first cooled to a temperature of 5^0C and then warmed back to a temperature of 20^0C. What can you conclude about the final states of the breakers?
   A) both breakers are full.
   B) both breakers are not full.
   C) only breaker-2 is full.
   D) breaker-1 contains more volume of water.

8. Metallic water pipes burst during cold seasons. This is because,
   A) the outer part of the pipe contracts more than the inner part.
   B) Metal contracts easily than water.
   C) Water expands when it freezes.
   D) Ice expands when it melts.

9. If a silver bar having a mass of 525g absorbs 6151 energy and its temperature rises by $50^0 C$c what is the heat capacity of silver?
   A) $234 J/^0 C$ B) $246 J/^0 C$ C) $0.234 J/^0 C$ D) $123 J/^0 C$

10. Heat is continuously supplied to a mass of water. The temperature (T) of the water varies with time (t) as shown in the figure below, what would happen during the time interval between t_1 and t_2? There is a phase transition from



   A) water to steam B) water to ice
   C) steam to ice D) steam to water

11. A square metal plate has area of $0.09 m^2$ at a temperature of $30^0 C$. if the temperature is raised to $60^0 C$ the area increases by (the coefficient of linear expansion for metal is $1.1 \times 10^{-5} /^0 C$)
   A) $1.000059 m^2$ B) $1.0001188 m^2$ C) $0.0000297 m^2$ D) $0.0000594 m^2 C$

12. How much heat energy is required to change an ice cube of mass $m = 720g$ from ice at a temperature of $T_i = -10^0 C$ to water at a temperature of $T_f = 15^0 C$? (Specific heat capacity of ice is $c_i = 2.220 \times 10^3 J/(kg.K)$, specific heat capacity of water is $c_w = 4.187 \times 10^3 J/kg.K$, and the latent heat of fusion of ice is $L_f = 3.330 \times 10^5 J/kg$
   A) $2.4 \times 10^4 J$ B) $6.12 \times 10^4 J$ C) $3.00 \times 10^5 J$ D) $2.68 \times 10^5 J$

13. The length of an aluminum rod at a temperature of 20^0C is 3m. What will be the ratio of change in the length of the rod to its original length if its temperature is raised to 120^0C? the coefficient of linear expansion of aluminum is $3.31x10^{-5}$/K

   A) 23% B) 3.2% C) 3% D) 0.33%

14. A physics student derived the formula for the surface expansion, ΔA, of a rectangular plate in terms of its linear expansion coefficient, α. Which of the following formulas is possibly derived by the student? (Assume that ΔT is the change in temperature of the plate and A_0 is its initial area.)

   A) $\Delta A = 2A_0 \alpha \Delta T$ C) $\Delta A = 3A_0 \alpha \Delta T$
   B) $\Delta A = \frac{1}{2} A_0 \alpha \Delta T$ D) $\Delta A = A_0 \alpha \Delta T$

15. A hot alloy with specific heat capacity of $200J/kg^0C$ cooled by 50^0C when added in to water. What amount of energy is given off by the alloy if its mass 0.3 kg?

    A) 3000J B) 5000J C) 1000J D) 1500J

16. A certain mass of ice at 00C is placed in a thermos bottle containing 0.1 kg of water at 6^0C. How many grams of ice will melt if the final temperature of the system is 0^0C? (The heat of fusion of water is $334,000J/Kg$ and the specific heat is $4,200J/kg^0C$)

    A) $1.7x10^{-3}kg$ B) $5.0x10^{-3}kg$ C) $7.5x10^{-3}kg$ D) $2.0x10^{-3}kg$